- training error : $J(\theta) = \frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - h_\theta(x^{(i)}))^2$
- test error : $L(\theta) = E_{(x,y)\sim P} (y - h_\theta(x))^2$

$\Rightarrow$ generalization gap

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underfitting : train ☹

overfitting : train ☺ test ☹

$f(x_0) + \varepsilon$ and $\hat{f}(x_0)$

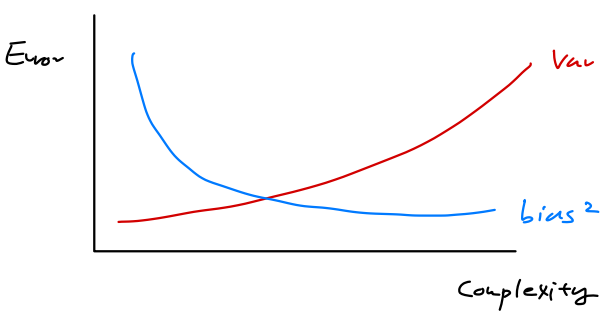
(i) $E[(f(x_0) - \hat{f}(x_0) + \varepsilon)^2] = E[(f(x_0) - \hat{f}(x_0))^2]$

$+ E[2\varepsilon (f(x_0) - \hat{f}(x_0))] + \underbrace{E[\varepsilon^2]}_{\text{irreducible}}$

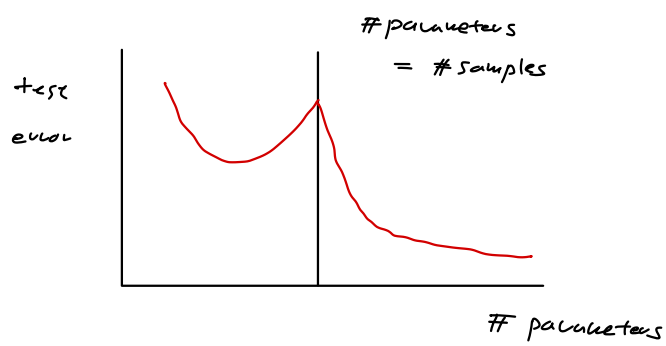
(ii) $E[(f(x_0) - \hat{f}(x_0))^2] = E[(f(x_0) - E[\hat{f}(x_0)])^2] + E[(E[\hat{f}(x_0)] - \hat{f}(x_0))^2]$

$= [\text{Bias}(\hat{f}(x_0))]^2 \qquad = \text{Var}(\hat{f}(x_0))$

Bias - Var Tradeoff



Double Descent Phenomenon



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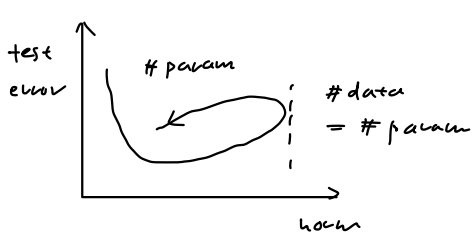
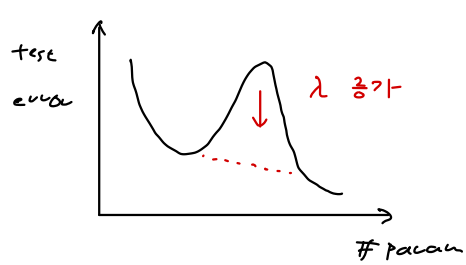
norm $\|\theta\|$ $\Rightarrow$ Double Descent Phenomenon $\Downarrow$



Linear Regression

$\Rightarrow X^T y = X^T X \theta + \lambda I$